MILITARY SPECIFICATION

OVEN ASSEMBLIES, WITH QUARTZ CRYSTAL UNITS, GENERAL SPECIFICATION FOR

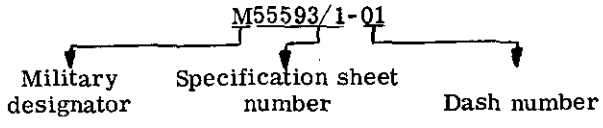
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the general requirements for non-repairable oven assemblies with integral quartz crystal units, designed for frequency control of electronic equipment.

1.2 Classification.

1.2.1 Military part number. The military part number shall consist of the letter "M", the basic number of the specification sheet, and an assigned dash number (see 3.1) as shown in the following:



2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Box, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-T-60	-	Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-R-10509	-	Resistors, Fixed, Film (High Stability) General Specification For.
MIL-C-45662	-	Calibration of Standards.

(See supplement 1 for list of associated specification sheets.)

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-147	-	Palletized and Containerized Unit Loads 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-456	-	Electronic Parts, Date and Source Coding for.

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:40Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

U. S. ARMY

TM 11-2652 - Crystal Impedance Meter TS-683/TSM.
TM 11-5051 - Crystal Impedance Meter TS-330/TSM.
TM 11-5106 - Quartz Crystal Unit Test Set TS-710/TSM.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 Preproduction. Oven assemblies, furnished under this specification, shall be a product which has been tested, and has passed the preproduction inspection tests specified in 4.4.

3.3 Materials. The materials shall be as specified (see 3.1). However, when materials are not specified, materials shall be used which will enable the oven assemblies to meet the requirements of this specification. When required by the contract or purchase order the supplier shall furnish a certification supported by verifiable data that materials used in fabricating the oven assemblies are in accordance with the specified requirements, and are the same materials used in fabricating the preproduction sample units.

3.4 Design and construction. Oven assemblies shall be of the design, construction, and physical dimensions specified (see 3.1).

3.4.1 Physical characteristics. Physical characteristics (such as weight, finish, color, etc.,) shall be as specified (see 3.1).

3.4.2 Heaters. Oven assemblies shall include built-in heaters as specified (see 3.1), and shall be capable of operation within performance requirements in any attitude or position. Heaters shall be wound in a manner to minimize inductance.

3.4.3 Terminals. Oven assembly terminals shall have the mechanical configuration and provide electrical connections as specified (see 3.1).

3.4.3.1 Pin alinement (when applicable). The pins in the base of the oven assembly shall freely and completely enter the specified pin alinement test gage (see 3.1).

3.4.3.2 Identification. Terminals shall be identified as specified (see 3.1).

3.5 Solderability (applicable to oven assemblies with wire leads). When oven assemblies are tested as specified in 4.6.2, the dipped surface of the lead shall be at least 95-percent covered with a continuous new solder coating. The remaining 5 percent may show only small pinholes or voids which shall not be concentrated in one area.

3.6 Capacitance. When oven assemblies are tested as specified in 4.6.3, the capacitance shall be as specified (see 3.1).

3.7 Frequency and equivalent resistance. The frequency and equivalent resistance of the oven assembly shall be within the limits specified when tested under the following conditions as applicable (see 3.1):

(a) Room ambient temperature (see 4.6.4.1).
(b) Operating temperature range (see 4.6.4.2).

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:40Z
Check the source to verify that this is the current version before use.

3.8 Frequency stability. Throughout the operating temperature range, the frequency of oven assemblies shall not deviate from the measured frequency at the room ambient temperature by more than the value specified (see 3.1), and shall also be within the specified frequency tolerance limits, when assemblies are tested as specified in 4.6.5.

3.9 Unwanted modes.

3.9.1 Method I. Unless otherwise specified (see 3.1), there shall be no unwanted modes of oscillation (resonant frequencies other than the desired operating frequency) when oven assemblies are tested as specified in 4.6.6.1. In addition, there shall be no abrupt frequency shifts and no intermittent oscillations. (Some assemblies may not start oscillating immediately at the plus 20 percent setting, or may cease oscillating during detuning, without resumption of oscillation on further detuning. These conditions are permitted.)

3.9.2 Method II (when specified (see 3.1)). When oven assemblies are tested as specified in 4.6.6.2, all unwanted modes shall have a resistance which exceeds 1.5 times the main mode resistance.

3.10 Insulation resistance. When oven assemblies are tested as specified in 4.6.7, the insulation resistance shall be 500 megohms, minimum.

3.11 Dielectric withstanding voltage.

3.11.1 At sea level. When oven assemblies are tested as specified in 4.6.8.1, there shall be no arcing, breakdown, or damage.

3.11.2 Barometric pressure (reduced) (applicable to oven assemblies for use above 10,000 feet altitude) (see 3.1). When oven assemblies are tested as specified in 4.6.8.2, there shall be no resultant impairment or damage sufficient to cause failure of the assemblies.

3.12 Warmup power. When oven assemblies are tested as specified in 4.6.9, the warmup power shall not exceed the value specified (see 3.1).

3.13 Warmup time. When oven assemblies are tested as specified in 4.6.10, the warmup time shall not exceed the value specified (see 3.1).

3.14 Average operating power. When oven assemblies are tested as specified in 4.6.11, the average operating power of the oven at -55° C (unless otherwise specified (see 3.1)), shall not exceed the value specified (see 3.1).

3.15 Shock. When oven assemblies are tested as specified in 4.6.12, changes in frequency and equivalent resistance of the oven assembly shall not exceed the values specified (see 3.1). There shall be no resultant impairment or damage sufficient to cause failure of the oven assembly.

3.16 Vibration. When ovens are tested as specified in 4.6.13, changes in frequency and equivalent resistance of the oven assembly shall not exceed the values specified (see 3.1). There shall be no resultant impairment or damage sufficient to cause failure of the oven assembly.

3.17 Seal (when applicable). The leakage rate of hermetically sealed assemblies shall not exceed 10^{-7} atmospheric cubic centimeters per second (atm cc/sec) when tested as specified in 4.6.14.

3.18 Thermal shock. When oven assemblies are tested as specified in 4.6.15, there shall be no resultant impairment or damage sufficient to cause failure of the assembly. The changes in frequency and equivalent resistance shall not exceed the value specified.

3.19 Salt spray (corrosion) (when applicable). There shall be no evidence of corrosion, and the frequency and equivalent resistance of the oven assemblies shall be within the limits specified in 3.7 after completion of the test specified in 4.6.16. Corrosion shall be considered excessive which causes impairment of the electrical or mechanical performance of the unit.

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:40Z
Check the source to verify that this is the current version before use.

3.20 Moisture resistance (when applicable). The frequency and equivalent resistance of the oven assemblies shall be within the limits specified in 3.7, after the test specified in 4.6.17 has been completed and the insulation resistance shall not be less than 500 megohms.

3.21 Life. When oven assemblies are tested as specified in 4.6.18, the difference between the highest and lowest frequencies measured shall not exceed the value specified (see 3.1).

3.22 Terminal strength.

3.22.1 Terminal pull. When oven assemblies are tested as specified in 4.6.19.1, there shall be no loosening, fracturing, or rupturing of the terminals; no other damage shall occur to the oven assemblies.

3.22.2 Wire-lead bend (applicable to oven assemblies with wire lead) (see 3.1). When oven assemblies are tested as specified in 4.6.19.2, there shall be no severing of the terminal.

3.23 Electromagnetic interference (EMI) (when applicable). When oven assemblies are tested as specified in 4.6.20, the level of transient interference measured during warm up and during the current cycling mode shall not exceed ±20 percent (peak voltage) of the applied line voltage unless otherwise specified (see 3.1).

3.24 Explosion (when applicable). When oven assemblies are tested as specified in 4.6.21, changes in frequency and equivalent resistance of the oven assembly shall not exceed the values specified (see 3.1).

3.25 Marking. The part number, specified frequency, and manufacturer's code designation shall be marked on the oven assembly in accordance with MIL-STD-130, at the locations specified (see 3.1). The part number, specified in 1.2.1, shall not be marked on any oven assembly for which the contractual requirements deviate in any way from the requirements specified herein for an assembly type. Any omitted, relaxed, added, or more stringent contractual requirement shall be considered a deviation. Oven assemblies shall also be marked with the year and week of the final test in accordance with MIL-STD-456; however, the manufacturer's code shall be by letters (see 6.5). Unless otherwise specified (see 6.2), no other marking shall be permitted on the oven assembly. The frequency shall be specified to the final hertz, and shall consist of 7 digits or less for frequencies below 10 megahertz (MHz), 8 digits for the range 10 MHz up to 100 MHz, and 9 digits for frequencies 100 MHz and above. Kilohertz (kHz) shall be used with frequencies below 1 MHz, and megahertz (MHz) with frequencies from 1 MHz and above. The frequency marking shall occupy a maximum of 13 spaces, as follows: a maximum of 9 digits; a space for a decimal point; and 3 spaces for "kHz" or "MHz", whichever is applicable; typical assembly markings are 833.333 kHz, 45.555556 MHz, and 123.888889 MHz. The part number, frequency, manufacturer's code designation characters shall be not less than 1/16 inch in height (unless otherwise specified), and located symetrically with respect to the center axis of the assembly.

3.26 Workmanship. The oven assemblies including all parts and accessories shall be manufactured and finished in a thoroughly workmanlike manner. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking, and plating. All dimensions and tolerances not specified shall be consistent with the best engineering practice. Where dimensions and tolerances affect the interchangeability, operation, or performance of the oven assembly they shall be held or limited accordingly. All material shall be sound, of uniform quality and condition, and free from seams, cracks, and other defects which may adversely affect the strength, endurance, or wear of the part. Any material hammered, filed, or treated in any other manner to conceal defects therein, shall be subject to immediate rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:40Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment and test facilities. Inspection equipment and test facilities shall be established and maintained in accordance with MIL-C-45662.

4.1.1.1 Supplier's equivalent test set. The supplier's equivalent test set shall be furnished by the supplier, and shall be used for the performance of quality conformance inspection. The characteristics of this set shall duplicate those of the applicable Government reference standard test set (see 4.1.2.1) and shall be correlated prior to performing quality conformance inspection. In addition, the supplier's equivalent test set shall be designed to permit external connections to the oven assembly, since it is not intended that the test set be subjected to the ambient temperature range specified for the oven assembly.

4.1.1.2 Frequency correlation. The frequency of a given oven assembly when oscillating in the supplier's equivalent test set (see 4.1.1.1) shall be within 0.0005 percent of its frequency when oscillating in the Government reference standard test set (see 4.1.2.1).

4.1.1.3 Resistance correlation. The equivalent resistance of a given oven assembly, as measured in the supplier's equivalent test set (see 4.1.1.1), shall be within 10 percent of its resistance as measured in the Government reference standard test set (see 4.1.2.1).

4.1.2 Test sets.

4.1.2.1 Government reference standard test sets. The Government reference standard test set shall be the specified crystal impedance meter (see 3.1), and shall be furnished by the Government at the supplier's plant. The Government reference standard test shall be used to check the characteristics and accuracy of the supplier's equivalent test set and shall not be used to perform quality conformance inspection. It shall be the supplier's responsibility to have the Government reference standard test set recertified at regular intervals of at least every six months and in accordance with paragraphs 3.2.3 and 3.2.5 of MIL-C-45662.

4.1.2.1.1 Procedure for adjustment and operation. The drive-level adjustment and operation of crystal impedance meters shall be in accordance with the applicable technical manual or other instructions, as follows:

Crytsal impedance meter	Technical manual
TS-330/TSM	TM 11-5051
TS-683/TSM and TS-683A/TSM	TM 11-2652
TS-710/TSM	TM 11-5106
AN/TSM-15	Applicable instruction book
AN/TSM-17	Applicable instruction book

Adjustment of CI meter frequency tuning control shall be in accordance with TM 11-5051 for TS-330/TSM and TM 11-2652 for TS-683A/TSM. The AN/TSM-15 test set shall be used in lieu of the TS-683/TSM test set for all frequencies above 100 MHz.

4.2 Classification of inspection. The examination and testing of oven assemblies shall be classified as follows:

(a) Preproduction inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:40Z
Check the source to verify that this is the current version before use.

4.3 Inspection conditions.

4.3.1 Conditions. Unless otherwise specified herein, all inspections shall be made in accordance with the general requirements of MIL-STD-202.

4.3.2 Measurements before and after a test. When frequency or equivalent resistance, or both, are measured before and after a test for determining change during the test, both measurements shall be made with the oven assembly in thermal equilibrium at the same temperature ±1 C, and the drive level set at the same practical minimum. Measured changes in frequency and equivalent resistance during the test shall not exceed the respective maximum changes specified for the particular test in question. However, at minimum drive level, it is not necessary that the unit operate within the overall tolerances for frequency and equivalent resistance since a change in the drive level does affect the operating characteristics of the crystal, especially its frequency. The oven shall be energized during tests where only "before and after" measurements are required, in order to minimize frequency shifts due to the thermal shock of cooling and heating the oven cavity.

4.3.3 Measurements after tests. When frequency or equivalent resistance or both, are to be measured after a test to determine compliance with specified tolerances, the measurements shall be made at the specified drive level. Measurements shall be made with the units in thermal equilibrium at the reference temperature.

4.3.4 Drive-level adjustment (see 4.6.4.2). When rated drive level is specified for a test, it shall be adjusted immediately before each test. No further adjustment shall be made between measurements during a temperature run test (except in case of line voltage fluctuation or a test lasting longer than an hour).

4.4 Preproduction inspection. Preproduction inspection shall be performed by the supplier, after award of contract and prior to production, at a laboratory acceptable to the Government. Preproduction inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. Preproduction approval is valid only on the contract, or purchase order under which it is granted, unless extended by the Government to other contracts or purchase orders. (See 6.2.)

4.4.1 Sample size. The number of oven assemblies to be submitted for preproduction inspection shall be in accordance with table I.

TABLE I. Preproduction sample sizes.

Order size	Sample size to be submitted
0-59	2
60-99	4
100-139	6
140-179	8
180 and over	10

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table II, in the order shown.

4.4.3 Failures. One or more failures shall be cause for rejecting the preproduction sample. In the event of failure and if the supplier is required to submit a new sample, the number shall be equal to the number of the original submission, and the new submission shall be accompanied by a description of the changes made to correct the cause of failure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:40Z
Check the source to verify that this is the current version before use.

TABLE II. Preproduction inspection.

Examination or test	Requirement paragraph	Test method paragraph
Visual and mechanical (external)	3.1, 3.3, 3.4, and 3.25	4.6.1
Solderability (when applicable)	3.5	4.6.2
Capacitance	3.6	4.6.3
Frequency and equivalent resistance:		
Room ambient temperature (when applicable)	3.7(a)	4.6.4.1
Operating temperature range (when applicable)	3.7(b)	4.6.4.2
Frequency stability	3.8	4.6.5
Unwanted modes	3.9	4.6.6
Insulation resistance	3.10	4.6.7
Dielectric withstanding voltage:		
At sea level	3.11.1	4.6.8.1
Barometric pressure (reduced) (when applicable)	3.11.2	4.6.8.2
Warm-up power	3.12	4.6.9
Warm-up time	3.13	4.6.10
Average operating power	3.14	4.6.11
Shock	3.15	4.6.12
Vibration	3.16	4.6.13
Seal (when applicable)	3.17	4.6.14
Thermal shock	3.18	4.6.15
Salt spray (corrosion) (when applicable)	3.19	4.6.16
Moisture resistance (when applicable)	3.20	4.6.17
Life	3.21	4.6.18
Terminal pull	3.22.1	4.6.19.1
Wire lead bend (when applicable)	3.22.2	4.6.19.2
Electromagnetic interference	3.23	4.6.20
Explosion (when applicable)	3.24	4.6.21

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection. Except as specified in 4.5.2.1.4 delivery of products which have passed the group A inspection shall not be delayed pending the results of the group B inspection.

4.5.1.1 Inspection lot. An inspection lot, shall consist of oven assemblies covered by a single specification sheet produced under essentially the same conditions and offered for inspection at one time.

4.5.1.1.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table III, in the order shown.

4.5.1.1.1.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:40Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection.

Examination or test	Requirement paragraph	Test method paragraph	AQL
Visual and mechanical (external)	3.1, 3.3, 3.4, and 3.25	4.6.1	1.0
Solderability (when applicable)	3.5	4.6.2	
Capacitance	3.6	4.6.3	1.5
Frequency and equivalent resistance:			
Room ambient temperature (when applicable)	3.7(a)	4.6.4.1	
Operating temperature range (when applicable)	3.7(b)	4.6.4.2	
Frequency stability	3.8	4.6.5	
Unwanted modes	3.9	4.6.6	
Insulation resistance	3.10	4.6.7	
Dielectric withstanding voltage:			
At sea level	3.11.1	4.6.8.1	
Barometric pressure (reduced) (when applicable)	3.11.2	4.6.8.2	
Warm-up power	3.12	4.6.9	
Warm-up time	3.13	4.6.10	
Seal (when applicable)	3.17	4.6.14	
Electromagnetic interference	3.23	4.6.20	
Explosion (when applicable)	3.24	4.6.21	

4.5.1.1.1.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.1.1.3 Disposition of sample oven assemblies. Sample units which have passed all the group A inspection may be delivered on the contract or purchase order, if the lot is accepted. No defective oven assemblies shall be shipped.

4.5.2 Periodic inspection. Periodic inspection shall consist of group B inspection.

4.5.2.1 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table IV, in the order shown. Group B inspection shall be made on sample units selected from inspection lots which have passed the group A inspection.

4.5.2.1.1 Sampling plan. The sample size shall be the same as for preproduction inspection (see table I). The sample units shall be selected from those covered by a single specification sheet, on the first lot produced after preproduction inspection and every 12 months thereafter. The sample units shall be divided equally into two groups and subjected to the tests of subgroups 1 and 2, respectively of table IV.

4.5.2.1.2 Failures. If one or more sample units fail to pass group B inspection, the sample shall be considered to have failed.

4.5.2.1.3 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:40Z
Check the source to verify that this is the current version before use.

TABLE IV. Group B inspection.

Examination or test	Requirement paragraph	Test method paragraph
Subgroup 1		
Capacitance	3.6	4.6.3
Insulation resistance	3.10	4.6.7
Dielectric withstanding voltage:		
At sea level	3.11.1	4.6.8.1
Barometric pressure reduced (when applicable)	3.11.2	4.6.8.2
Average operating power	3.14	4.6.11
Shock	3.15	4.6.12
Vibration	3.16	4.6.13
Seal (when applicable)	3.17	4.6.14
Thermal shock	3.18	4.6.15
Salt spray (corrosion) (when applicable)	3.19	4.6.16
Moisture resistance (when applicable)	3.20	4.6.17
Terminal pull	3.22.1	4.6.19.1
Wire-lead bend (when applicable)	3.22.2	4.6.19.2
Electromagnetic interference	3.23	4.6.20
Explosion (when applicable)	3.24	4.6.21
Subgroup 2		
Life	3.21	4.6.18

4.5.2.1.4 Noncompliance. If a sample fails to pass group B inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Group A inspection may be reinstituted; however, final acceptance shall be withheld until the group B reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.5.2.2 Final test. Each individual oven assembly on a prime Government contract or order shall have been subjected, within 15 days prior to date of shipment, to test for frequency as applicable (see 4.6.4). No oven assembly failing this test shall be delivered.

4.5.3 Inspection of preparation for delivery. Sample packages or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 or the documents specified herein.

4.6. Methods of examination and test.

4.6.1 Visual and mechanical examination. Oven assemblies shall be examined to verify that the materials, design, construction, physical dimensions, and markings, are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, and 3.25).

4.6.2 Solderability (applicable to oven assemblies with wire leads) (see 3.5). Each wire lead terminal shall be subjected to method 208 of MIL-STD-202.

Source: https://assist.dla.mil -- Downloaded: 202640-22T23:40Z
Check the source to verify that this is the current version before use.

4.6.3 Capacitance (see 3.6). Oven assemblies shall be tested in accordance with method 305 of MIL-STD-202. The capacitance shall be measured from pin to pin, with the assembly grounded, at a frequency which is lower than the fundamental frequency of the assembly, and at which the assembly shows no oscillation response.

4.6.4 Frequency and equivalent resistance.

4.6.4.1 At room ambient temperature (see 3.7(a)). With the oven assembly operating in the applicable test set (see 3.1 and 4.1.2.1.1) or a suitable correlated oscillator and in thermal equilibrium at the specified room ambient temperature, the frequency and the equivalent resistance shall be measured.

4.6.4.2 Operating temperature range (see 3.7(b)). Measurements of frequency and equivalent resistance of an oven assembly shall be performed at rated drive level (see 4.3.4), and shall be monitored continuously over the operating temperature range. The rate of temperature change shall not exceed 2° C per minute.

4.6.5 Frequency stability (see 3.8). Oven assemblies shall be subjected to a temperature run over the operating range (see 3.1). The rate of change of temperature shall not exceed 25° C per minute. The units shall be monitored for frequency and equivalent resistance.

4.6.6 Unwanted modes.

4.6.6.1 Method I (unless otherwise specified) (see 3.1 and 3.9.1). Oven assemblies shall be tested at rated drive level (see 3.1 and 4.3.3) with the oven assembly in thermal equilibrium at the room ambient temperature using the following procedure:

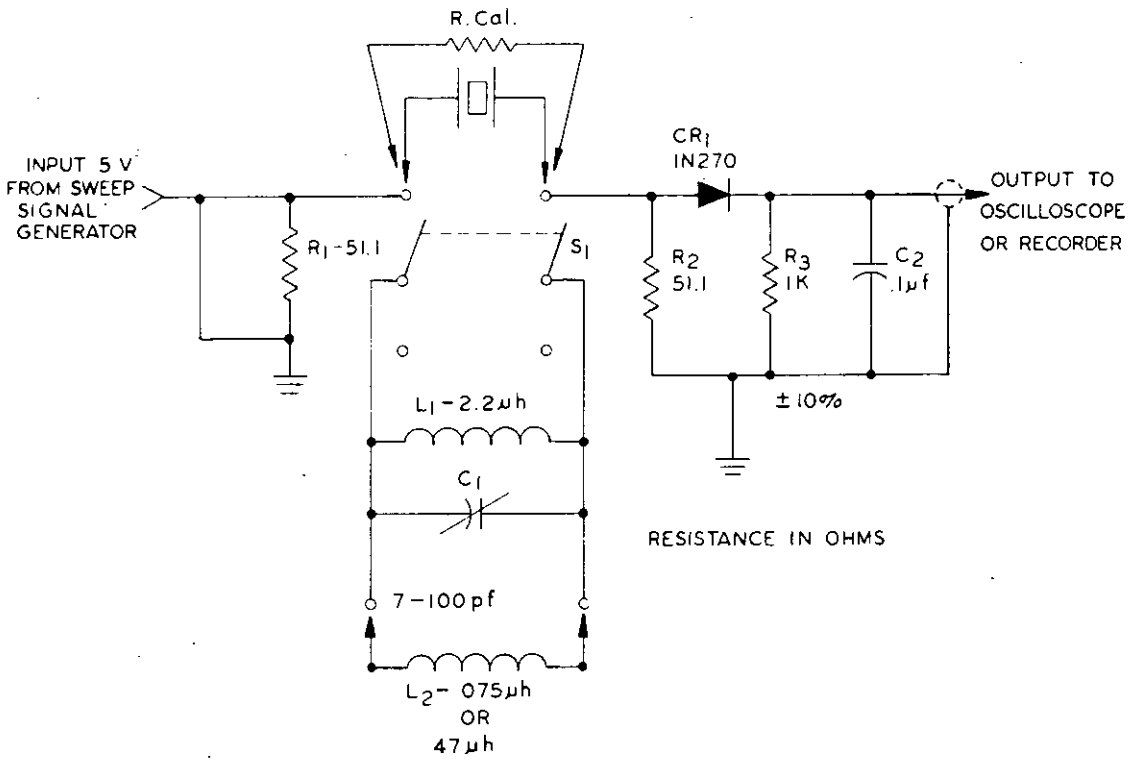
(a) Determine the value of maximum equivalent resistance (see 3.7). Insert in the test set a fixed resistor equal to this value. Adjust the output frequency of the test set to a frequency 20 percent lower than the specified frequency, and then to a frequency 20 percent higher than the specified frequency. The adjacent band of the test set, or another test set shall be used when necessary.
(b) Record the dial setting of the tuning control on the test set obtained in (a).
(c) Replace the fixed resistor with the oven assembly. The test set should be at the plus 20 percent frequency setting before inserting the assembly.
(d) Slowly vary the tuning control between the dial settings recorded in (b), beginning with the setting 20 percent higher than the specified frequency, while monitoring the output signal.

4.6.6.2 Method II (when specified) (see 3.1 and 3.9.2). Oven assemblies shall be tested using the test circuit in figure 1. The test shall be performed at room ambient temperature for all assemblies. A frequency range of approximately 200 kHz above 200 kHz below the specified frequency shall be swept by the signal generator. S_1 shall be set in the open position for assemblies with operating frequencies below 12 MHz. For assemblies with operating frequencies above 12 MHz, S_1 shall be set in the closed position, the proper inductance value for L_2 shall be used, and C_1 shall be adjusted to parallel resonate the static capacitance (C_O), as evidenced by an indication of zero or minimum amplitude on the oscilloscope or recorder. The output oscilloscope or recorder shall be observed for unwanted modes. All unwanted modes, whose resistance does not exceed one and a half times the main mode resistance shall be considered a failure.

4.6.7 Insulation resistance (see 3.10). Oven assemblies shall be tested in accordance with method 302 of MIL-STD-202. The following details shall apply:

(a) Test-condition letter - B.
(b) Points of measurement - Between all grounded terminals connected together, and all ungrounded terminals connected together, except that for assemblies using grounded power sources they shall be as specified (see 3.1).
(c) Electrification time - One minute.

Source: https://assist.dla.mil -- Downloaded: 202640-22T23:40Z
Check the source to verify that this is the current version before use.



NOTE: R_1 and R_2 shall be a 1% film resistor in accordance with MIL-R-10509.

FIGURE 1. Unwanted modes test circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:40Z
Check the source to verify that this is the current version before use.

4.6.8 Dielectric withstanding voltage.

4.6.8.1 At sea level (see 3.11.1). Oven assemblies shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

(a) Magnitude of test voltage - 500 volts, rms for preproduction inspection; for quality conformance inspection, test voltage shall be 600 volts, rms for one second.
(b) Nature of potential - AC.
(c) Points of application of test voltage - as specified in 4.6.7(b).

4.6.8.2 Barometric pressure (reduced) (applicable to assemblies for use above 10,000 feet altitude) (see 3.1 and 3.11.2). Oven assemblies shall be tested in accordance with 4.6.8.1 and method 105 of MIL-STD-202. The following details shall apply:

(a) Method of mounting - By normal mounting means in the normal position and attitude.
(b) Test-condition letter - C.
(c) Magnitude of test voltage - 200 volts, rms for one minute.
(d) Exposure time prior to measurements - One hour.

Following this test, assemblies shall be examined for impairment or damage, which would cause failure of the assemblies.

4.6.9 Warmup power (see 3.12). Oven assemblies, nonoperating, shall be subjected to an ambient temperature of -55° C (unless otherwise specified (see 3.1)), for a period of 3 hours. While the assemblies remain in this ambient temperature, the rated voltage (see 3.1) shall be applied and the warmup power to the assembly heaters shall be measured immediately.

4.6.10 Warmup time (see 3.13). Oven assemblies, nonoperating, shall be subjected to an ambient temperature of -55° C (unless otherwise specified (see 3.1)) for a minimum period of 3 hours. While the ovens remain in this ambient temperature, rated voltage (see 3.1) shall be applied and the elapsed time shall be measured from the instant of application of voltage to the time when the oven assembly comes to and remains within the allowable tolerance of the specified frequency.

4.6.11 Average operating power (see 3.14). The measurement for average operating power shall be made at -55° C ambient extreme (unless otherwise specified (see 3.1)), after the specified warmup time (see 3.1). Where cyclic thermostatic control is used, this measurement is determined by measuring the peak power while the operating thermostat is closed, the ON time, and the OFF time. The average power after stabilization is then calculated using the formula:

$$\frac{\text{ON time x peak power}}{\text{ON time + OFF time}} = \text{Average operating power}$$

4.6.12 Shock (see 3.15). Oven assemblies shall be tested in accordance with method 213 of MIL-STD-202. The following details shall apply.

(a) Test condition - I (100G).
(b) Measurements immediately before and after test - Frequency and equivalent resistance in accordance with 4.6.4.
(c) The oven assembly shall be securely fastened to the shock test platform.
(d) The oven assembly shall not be energized.

4.6.13 Vibration (see 3.16). Oven assemblies shall be energized (with crystal oscillating) and shall be tested in accordance with MIL-STD-202, using the method specified (see 3.1). The following details and exceptions shall apply:

(a) Tests and measurements immediately prior to vibration - The frequency and equivalent resistance shall be measured as specified in 4.3.2 and 4.6.4.

Source: https://assist.dla.mil Downloaded: 2026-10-22T23:40Z
Check the source to verify that this is the current version before use.

(b) Test methods as specified - (see 3.1):
Method 201 - 2 hours or
Method 204 - Test condition A, 3 hours.
(c) Method of mounting and direction of motion - Assemblies shall be rigidly mounted on the horizontal platform of a vibration machine in such a manner that the applied vibration shall be as follows:
1. One-third of the assemblies (to the nearest integral number) shall have the direction of vibration parallel to the pin length.
2. The same number of assemblies shall have the direction of vibration perpendicular to the largest surface.
3. The remainder of the sample unit assemblies shall have the direction of vibration perpendicular to the pin length and parallel to the largest surface.
(d) Time of traverse of frequency range - 1 to 2 minutes (method 201).
(e) Tests and measurements immediately after vibration - The frequency and equivalent resistance shall be measured as specified in 4.3.2 and 4.6.4.

4.6.14 Seal (when applicable) (see 3.17). Hermetically sealed oven assemblies shall be tested in accordance with method 112 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test-condition letter - C.
(b) Procedure - III for evacuated enclosure; and IV, for backfilled enclosure.
(c) Leakage rate sensitivity - 10^{-7} atm cc/sec.

4.6.15 Thermal shock (see 3.18). Oven assemblies shall be tested in accordance with method 107, MIL-STD-202. Measurements shall be made before and after the tests in accordance with 4.3.2 and 4.6.4. The following details and exceptions shall apply:

(a) Test-condition letter - A.
(b) Measurements after cycling - Assemblies shall be tested as specified in 4.6.4.

Following the test, assemblies shall be examined for impairment or damage which would cause failure of the assemblies.

4.6.16 Salt spray (corrosion) (when applicable) (see 3.19). Oven assemblies shall be tested in accordance with method 101, test condition B, of MIL-STD-202. After this test, the unit shall be visually examined for evidence of corrosion, and the frequency and equivalent resistance shall be measured as specified in 4.3.1, 4.3.3, and 4.6.4.

4.6.17 Moisture resistance (when applicable) (see 3.20). Oven assemblies shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting - Normal mounting means, except during measurements.
(b) Initial measurements - Not applicable.
(c) Subcycle - Step 7b, the vibration subcycle, shall be omitted.
(d) Polarization and loading voltage - Not applicable.
(e) Final measurement - After the drying period, frequency and equivalent resistance shall be measured as specified in 4.6.4, and insulation resistance shall be measured in accordance with method 302 of MIL-STD-202, using a test potential of 50 to 55 volts. Insulation resistance shall be measured from terminal to terminal and from each terminal to the assembly case.

4.6.18 Life (see 3.21). Oven assemblies shall be energized (with crystal oscillating) at room ambient temperature for a continuous period of thirty days. The frequency shall be measured twice a week at intervals of not less than two days or more than four days. The initial measurement of frequency shall be taken at the end of the first forty-eight hours, and the final measurement at the end of the thirty-day test. All oven assemblies (including those designed for anti-resonance operation) shall be measured at series resonance whenever crystal impedance meters are utilized (see 4.1.2). When other test oscillators are employed, the oven assembly may be operated above the series resonant frequency. The drive level shall be as specified in 4.3.2. The same test set shall

Source: https://assist.dla.mil -- Downloaded: 20260-22T23:40Z
Check the source to verify that this is the current version before use.

be used throughout the test. In the event of any condition which causes the oven assembly to be de-energized for a time interval of more than one hour, no measurement shall be made until forty-eight hours after the assembly has been re-energized. The thirty-day test period shall be extended by the length of time the test has been interrupted. The resolution of the measuring system shall be ten times or greater than the peak to peak specified frequency division over a thirty day period.

4.6.19 Terminal strength.

4.6.19.1 Terminal pull (see 3.22.1). Unless otherwise specified (see 3.1), oven assemblies shall be tested in accordance with method 211 of MIL-STD-202. The following details shall apply:

(a) Test condition - A.
(b) Applied force - 4-1/2 pounds.

4.6.19.2 Wire lead bend (applicable to oven assemblies with wire leads) (see 3.1 and 3.22.2). Oven assemblies shall be tested in accordance with method 211 of MIL-STD-202. The following details shall apply:

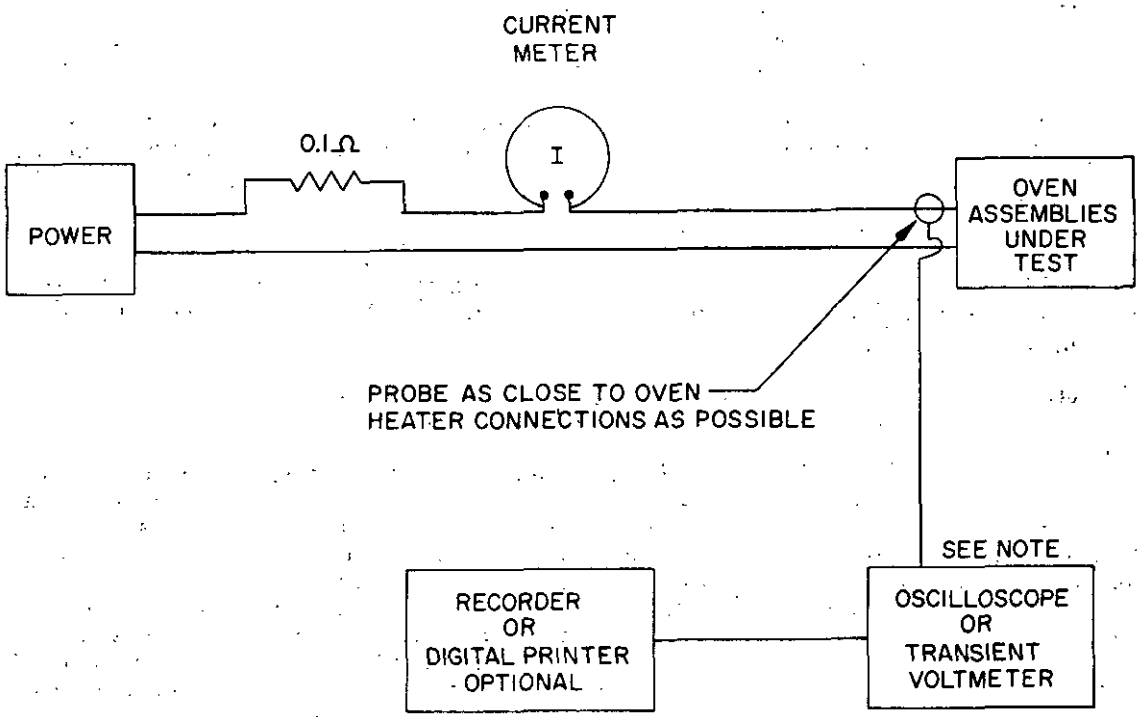
(a) Test condition - C.
(b) Applied load - 1 pound.

4.6.20 Electromagnetic interference (EMI) (see 3.23). The oven assembly shall be tested for transient interference at room ambient temperature in the test circuit as shown in figure 2. Input power to the oven heater element shall be at maximum voltage and shall not exceed the specified limits. The oven shall be monitored for noise continuously from initiation of input power to the oven heater element and during the time in which the oven is in its current cycling mode as indicated by the current meter in the power line. During the current cycling mode, monitoring for noise shall be continued for a time sufficient to indicate that no significant increase in noise is generated by the oven under test. The highest reading obtained shall be recorded. Means to accelerate the number of cycles during the current cycling mode after warmuup may be applied by providing a low environmental temperature to the oven. The low temperature environment shall not exceed the -55° C limit, unless otherwise specified (see 3.1).

4.6.21 Explosion (when applicable) (see 3.24). Oven assemblies shall be tested in accordance with method 109 of MIL-STD-202. The following detail shall apply:

(a) Measurements before and after test - Frequency and equivalent resistance shall be measured as specified in 4.6.4.
(b) Mechanical and electrical load - Oven assemblies shall be operated at their maximum rated dc inductive current.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:40Z
Check the source to verify that this is the current version before use.



NOTE:
Oscilloscope or transient voltmeter shall be of the memory storage type and be capable of detecting and measuring:

Magnitude: ±1 volt to ±1000 volts
Width: 50 nanoseconds to 100 milliseconds
PRF: Single shot to 500 PPS
Input impedance: 100k ohms min

Figure 2. Electromagnetic interference test circuit.

Source: https://assist.dla.mil -- Downloaded: 20264.0-22T23:40Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Oven assemblies shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Oven assemblies shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packaging. Unless otherwise specified (see 6.2), oven assemblies shall be individually packaged in accordance with MIL-P-116, submethod IC-2 insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein.

5.1.1.5 Intermediate packaging. Oven assemblies, packaged as described in 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size, shape and quantities, shall be of minimum tare and cube and shall contain multiples of five unit packages, not to exceed 50 packages. No intermediate packaging is required when the total quantity shipped to a single destination is less than 50 unit packages.

5.1.2 Level C. Cleaned and dried oven assemblies shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. The packaged oven assemblies shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joints with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B. The packaged oven assemblies shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto. For Army procurement, fiberboard containers shall be class weather-resistant as specified in level A.

5.2.3 Level C. The packaged oven assemblies shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Oven assemblies, packed as specified in 5.2.1, shall be placed on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:40Z
Check the source to verify that this is the current version before use.

5.2.4.2 Level B. Oven assemblies, packed as specified in 5.2.2, shall be palletized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic. For Army procurement, caps shall be weather resistant as specified in level A.

5.2.4.3 Level C. Oven assemblies, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Navy procurements. For Navy procurements the use of polystyrene loose fill material (such as strips, strands and beads) is prohibited for packaging and packing applications.

6. NOTES

6.1 Preproduction. Information pertaining to preproduction inspection of products covered by this specification shall be obtained from the procuring activity issuing the contract or purchase order.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet, and the part number (see 1.2.1 and 3.1).
(c) Frequency or frequencies required.
(d) Any additional marking required on oven assembly (see 3.25).
(e) Levels of preservation and packaging and packing, and applicable marking (see section 5).

6.2.1 Indirect shipments. The preservation and packaging, packing and marking specified in section 5 apply only to direct purchases by or direct shipments to the Government and are not intended to apply to contracts or orders between the supplier and prime contractor.

6.3 Frequency range. Crystal units applicable to this specification cover a frequency range from 2.75 kilohertz to 134.0 megahertz with variations in holder, frequency tolerance, frequency stability, mode of operation etc.

6.4 Definitions. Refer to MIL-C-3098 and MIL-O-39021 for pertinent definitions.

6.5 Manufacturer's code. Manufacturer's code is listed in publication NAVSHIPS 0967-190-4010, "Manufacturer's Designating Symbols."

6.6 Additional lead length. Where additional lead length to the oven assembly during a test is required, the frequency and effective resistance at room ambient temperature shall be determined with that additional lead length and may not be within the specification tolerance.

Source: https://assist.dla.mil -- Downloaded: 202640-22T23:40Z
Check the source to verify that this is the current version before use.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MI, MU, SM
Navy -
Air Force - 11, 85
DSA - ES

User activities:
Army -
Navy - SH, AS, OS, CG, MC
Air Force - 17, 19

Preparing activity:
Army - EL

(Project 5955-0329)

Source: https://assist.dla.mil -- Downloaded: 202640-22T23:40Z
Check the source to verify that this is the current version before use.